telnyx

15 measured voices, first audio ranked, RTF added

TTS LATENCY BENCHMARK ACROSS MODEL FAMILIES

We reran the benchmark with RTF enabled and added Rime Coda. The primary comparison is first audio latency. Final audio is no longer used as a ranking metric because output duration and chunking behavior differ by provider.

FASTEST FIRST AUDIO P50

302 ms

Amazon Polly Standard, Joanna had the lowest first audio p50 in this run. Every measured voice completed 6/6 runs.

WEBSOCKET
Polly Standard, Joanna
302 ms
first audio p50

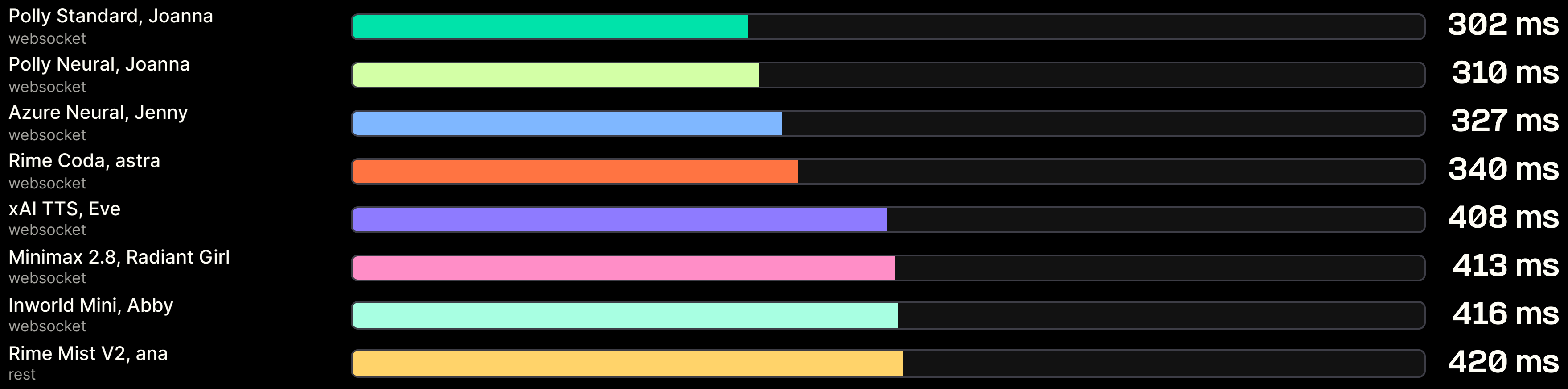
WEBSOCKET
Polly Neural, Joanna
310 ms
first audio p50

WEBSOCKET
Azure Neural, Jenny
327 ms
first audio p50

WEBSOCKET
Rime Coda, astra
340 ms
first audio p50

WEBSOCKET
xAI TTS, Eve
408 ms
first audio p50

WEBSOCKET
Minimax 2.8, Radiant Girl
413 ms
first audio p50



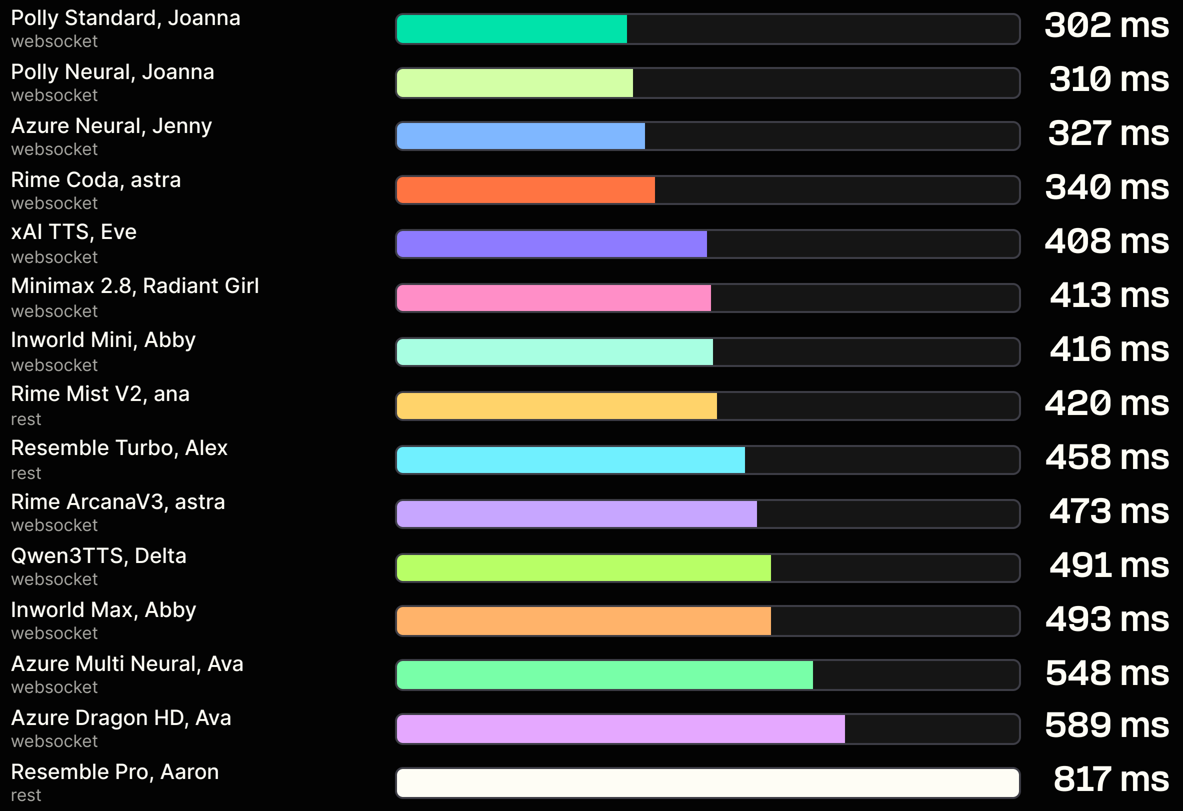
Prompt set: greeting, routing, and short agent response

Benchmark run, May 26, 2026

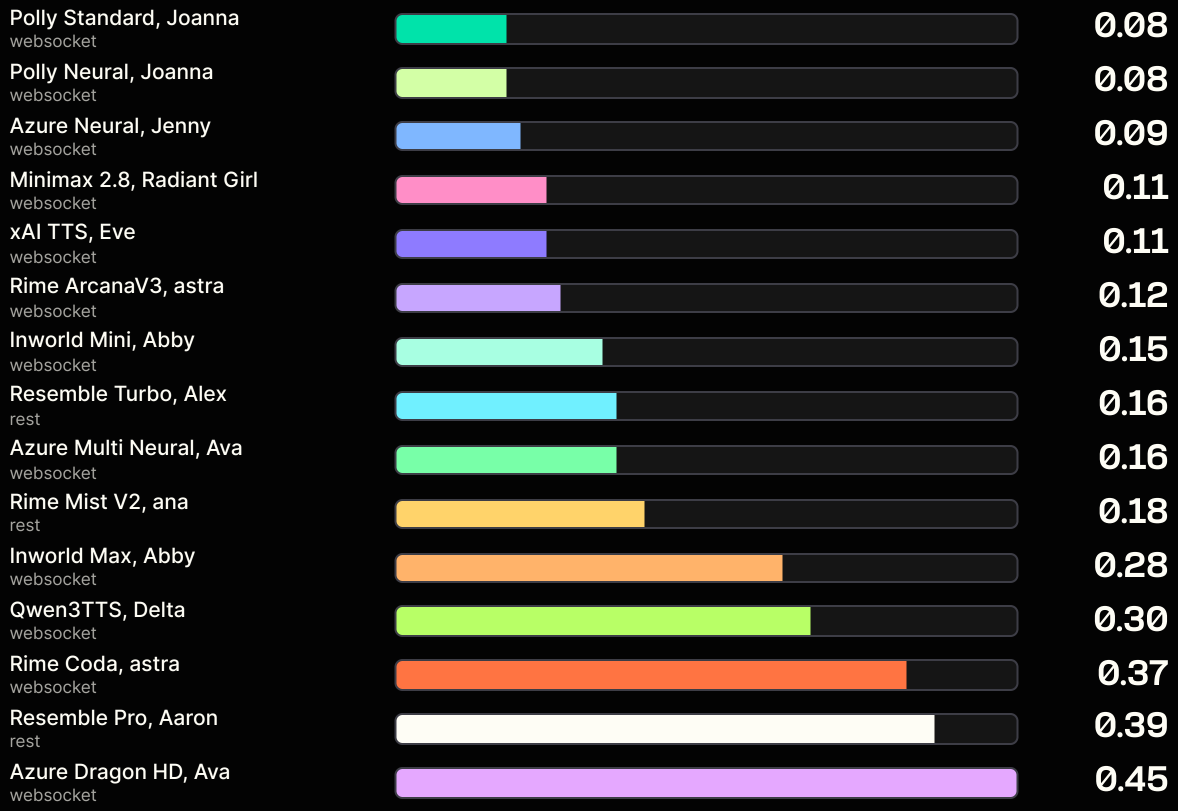
RTF REPLACES FINAL AUDIO RANKING

RTF normalizes synthesis time by generated audio duration. Lower is better. This is a better synthesis-efficiency comparison than final audio latency because voices speak at different speeds.

First audio p50



RTF p50



RTF = final observed audio time divided by generated audio duration

Duration source: MP3 metadata, ffprobe MP3, Qwen3TTS PCM bytes, or linear16 bytes

WHAT THE RUN SAYS

All measured voices returned successful audio for 6/6 runs. First audio is the cleanest cross-provider latency signal. RTF is included for synthesis efficiency. Final audio is kept out of ranking because chunking and voice speed differ.

FASTEST START

Polly Standard leads first audio

Amazon Polly Standard, Joanna finished at 302 ms first audio p50.

NEW ROW

Rime Coda is included

Rime Coda, astra finished at 340 ms first audio p50 and 0.37 RTF p50.

METHODOLOGY FIX

Final audio is not ranked

Single-chunk providers can have first audio equal final audio, so final audio is not a fair ranking metric.

MODEL FAMILY AND VOICE	INTERFACE	FIRST AUDIO P50	FIRST AUDIO P95	RTF P50	RTF P95	SUCCESS
Amazon Polly Standard, Joanna	websocket	302 ms	335 ms	0.08	0.09	6/6
Amazon Polly Neural, Joanna	websocket	310 ms	580 ms	0.08	0.17	6/6
Azure Neural, Jenny	websocket	327 ms	629 ms	0.09	0.12	6/6
Rime Coda, astra	websocket	340 ms	414 ms	0.37	0.41	6/6
xAI TTS, Eve	websocket	408 ms	1037 ms	0.11	0.56	6/6
Minimax 2.8 Turbo, Radiant Girl	websocket	413 ms	686 ms	0.11	0.23	6/6
Inworld Mini, Abby	websocket	416 ms	567 ms	0.15	0.21	6/6
Rime Mist V2 via Natural, ana	rest	420 ms	443 ms	0.18	0.21	6/6
Resemble Turbo, Alex	rest	458 ms	618 ms	0.16	0.32	6/6
Rime ArcanaV3, astra	websocket	473 ms	844 ms	0.12	0.63	6/6
Qwen3TTS, Delta	websocket	491 ms	589 ms	0.30	0.48	6/6
Inworld Max, Abby	websocket	493 ms	616 ms	0.28	0.31	6/6
Azure Multilingual Neural, Ava	websocket	548 ms	637 ms	0.16	0.32	6/6
Azure Dragon HD, Ava	websocket	589 ms	786 ms	0.45	0.54	6/6
Resemble Pro, Aaron	rest	817 ms	832 ms	0.39	0.41	6/6

COVERAGE MAP

This page maps model-family names to the representative voice IDs used in the benchmark. Ultra and Google were not included in the measured set.

REQUESTED MODEL FAMILY	REPRESENTATIVE TESTED	STATUS	PATH
Minimax 2.8 Turbo	Minimax 2.8 Turbo, Radiant Girl	Measured	WebSocket
Rime ArcanaV3	Rime ArcanaV3, astra	Measured	WebSocket
Rime Coda	Rime Coda, astra	Measured	WebSocket
Rime Mist V2	Rime Mist V2 via Natural, ana	Measured	REST
Inworld Mini	Inworld Mini, Abby	Measured	WebSocket
Inworld Max	Inworld Max, Abby	Measured	WebSocket
Qwen3TTS	Qwen3TTS, Delta	Measured	WebSocket
xAI TTS	xAI TTS, Eve	Measured	WebSocket
Resemble	Resemble Turbo and Pro	Measured	REST
Amazon Polly Standard	Amazon Polly Standard, Joanna	Measured	WebSocket
Amazon Polly Neural	Amazon Polly Neural, Joanna	Measured	WebSocket
Azure Neural	Azure Neural, Jenny	Measured	WebSocket
Azure Multilingual Neural	Azure Multilingual Neural, Ava	Measured	WebSocket
Azure Dragon HD	Azure Dragon HD, Ava	Measured	WebSocket
Telnyx Ultra	Not included	Not measured	Not accessible in this pass
Google Neural	No voice ID in live API	Not measured	Not exposed by Voices API
Google Chirp 3 HD	No voice ID in live API	Not measured	Not exposed by Voices API
Azure Standard	No standard voice in live API	Not measured	Only Neural and Dragon HD surfaced

Source: live Telnyx Voices API snapshot

Representative English voices were selected where possible.

HOW TO READ THIS BENCHMARK

The benchmark sends the same prompt set to each representative voice and records when audio first arrives. RTF is calculated when audio duration can be parsed or inferred.

MAIN METRIC

First audio p50

Median time from sending text to receiving first audio bytes.

SYNTHESIS EFFICIENCY

RTF p50

Median synthesis time divided by generated audio duration.

NOT RANKED

Final audio

Kept out of the ranking because chunking and voice speed differ by provider.

TRANSPORT

Interface

WebSocket for most providers. REST for Natural and Resemble in this run.

RUN COUNT

6 per voice

Three prompts with two runs per prompt. All measured voices completed 6/6.

ACCESS NOTE

Ultra and Google

Ultra was left out based on access. Google voices were not visible in the API.

WHAT THIS MEANS FOR USE CASES

Use this as a decision aid, not a universal voice ranking. TTS choice still depends on voice quality, language, cloning needs, transport, and playback strategy.

LIVE VOICE AGENTS

Watch first audio

This is the delay users feel before the assistant starts speaking.

LONG CLIPS

Watch RTF

RTF is the better normalized synthesis metric for longer generated responses.

REST CAVEAT

Separate transport

REST responses can stream chunks, but they are not the same path as interactive WebSocket text streaming.

PROVIDER CAVEAT

Single chunk

Polly and some others can return all audio at once, so final audio equals first audio.

SALES CAVEAT

Do not overclaim

Customer text and target voices should be tested before final choice.

NEXT RUN

Region control

Use the same client region, concurrency, prompts, and endpoint for comparisons.

Source: model-coverage-tts-benchmark-rtf-coda-results.json

Region: local macOS client, concurrency 1, mp3 output